

CLEAN CORE: AI-DRIVEN WASTE MONITORING AND OPTIMIZATION

Final Year Project (FYP) Defense Addendum

Prepared for: Internal Review Committee

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Subject: Additional Context Regarding Project Scope, Logistics Methodology, and Dataset Provenance

We sincerely appreciate the committee's thorough review and valuable feedback on our project. Your insights highlight important considerations for a comprehensive waste management system. To build upon your comments, this document aims to provide additional context regarding our specific design choices, the constraints we navigated with our industry partner, and the proprietary nature of the hardware utilized in our Final Year Project (FYP).

1. Clarification on Routing Scope: Fleet Management vs. Worker-Level Optimization

Addressing the Committee's Insight on Fleet Management:

The committee raised a highly relevant point regarding the need to track truck capacity, handle mid-route full trucks, define start/end depots, and calculate the total number of trucks. We entirely agree that a macro-level fleet management system is essential for city-wide logistics.

Our Project Focus and Industry Context:

During the scoping phase of **Clean Core**, we held extensive consultations with our industry partner, whose CEO has nine years of experience in the European waste management market. They advised us that macro-level fleet management is typically handled by established commercial logistics APIs (such as HERE Maps, Google, or DHL) that enterprise systems integrate with directly.

Therefore, rather than replicating these existing macro-level systems, we intentionally narrowed the scope of our FYP to tackle the micro-level algorithmic challenge. We focused on highly optimized **worker-level routing**. By framing the immediate collection route as a variant of the Travelling Salesman Problem (TSP), we deployed the Greedy Nearest Neighbor algorithm combined with 2-opt optimization. We hope the committee views this as a mathematically sound, localized routing solution designed to act as an intelligent layer that can eventually feed into the larger fleet management networks you rightly identified.

2. Hardware Constraints and Data Transmission

Addressing the Committee's Suggestion for Real-Time Image Transmission:

The committee made an excellent suggestion regarding the transmission of real-time, live images from the deployment sites in Belgium to the university campus to demonstrate live functionality. We thoroughly evaluated this approach during development, as real-time feedback is always the ideal scenario.

Technical and Hardware Realities:

We would like to clarify the specific nature of the hardware provided by our industry partner. The devices installed on the waste containers are specialized, low-cost (150–200 Euros), battery-operated IoT image sensors, rather than continuous CCTV cameras.

Because these sensors operate entirely on internal batteries and transmit via cellular mobile data networks, maintaining a continuous or on-demand live feed presents significant technical challenges. Doing so would rapidly deplete the hardware's battery life and incur substantial international cloud and bandwidth costs, which our industry partner noted as a primary limitation in real-world deployments.

Our Interval-Based Approach:

To align with industry standards for battery preservation and data efficiency, the hardware is engineered to capture automated, timestamped static snapshots at scheduled intervals. We found that this interval-based approach provides sufficient, actionable data for the **Clean Core** system to successfully:

- Calculate fill levels (from "Empty" to "Overflowing").
- Classify waste composition (e.g., Glass, Metal, Paper, Plastic, Organic).
- Detect contamination and illegal/hazardous materials (e.g., lithium-ion batteries, medical waste).

3. The Proprietary Nature of the Dataset

Finally, we would like to highlight the unique value of the dataset utilized for this FYP. Acquiring localized, real-world waste imagery independently is restricted due to privacy and operational laws at European container parks.

Thanks to our industry partner, we were granted access to a highly restricted, proprietary dataset gathered under explicit, legally binding agreements. We utilized these exclusive, timestamped sensor snapshots to fine-tune our custom YOLOv8 AI models via transfer learning. This allowed us to train our AI to meet strict, real-world environmental requirements (such as the DEFRA directive) using specialized data that is not available in generic, public AI training corpora.

Conclusion:

We are incredibly grateful for the committee's time and feedback, which helped us critically evaluate our work. We hope this additional context clarifies how industry constraints and hardware limitations shaped our focus on localized algorithmic optimization and specialized waste classification for the **Clean Core** project.